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Discontinuous finite element solution of transport equations for high-frequency power flows in slender structures

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ABSTRACT: This paper deals with some recent developments for the modeling and numerical simulation of high-frequency (HF) vibrations of heterogeneous slender structures. A time-domain approach is proposed to take account of the multiple reflection and scattering of vibrational waves in this frequency range. The mathematical model is derived from the semiclassical analysis of strongly oscillating (HF) solutions of quantum and classical wave systems, including acoustic, electromagnetic, or in the present case elastic waves. This theory shows that the associated phase space energy density satisfies a transport equation in the HF range. The proposed model also considers energetic boundary and interface conditions consistent with the boundary and interface conditions imposed to the solutions of the wave system. They are given in the form of power flow reflection/transmission operators for the energy rays impinging on a boundary or an interface. Nodal/spectral discontinuous "Galerkin" finite element methods are implemented to integrate the transport equations supplemented with boundary and interface conditions. Some numerical simulations are presented to illustrate the theory: the first one deals with an assembly of thick beams, and the second one with a flat plate. This research applies to the prediction of the linear transient responses of complex structures to impact loads or shocks, or the analysis of non-destructive evaluation techniques.

KEY WORDS: High frequency; Vibration; Transport; Discontinuous finite elements.

1 INTRODUCTION

Engineering structures exhibit typical transport and diffusive behaviors in the higher frequency (HF) range of vibration [1, 2]. These regimes are described by linear transport equations for the energy density associated to the strongly oscillating (HF) solutions of the elastic wave equation. More generally this result holds for all symmetric hyperbolic systems, including quantum waves and classical acoustic or electromagnetic waves [3–6]. It has been specialized to slender visco-elastic structures, typically beams, plates and shells in [7–9] for applications to aerospace structures. The mathematical model is derived from the semiclassical analysis of HF solutions of such wave systems: it shows that the associated energy density in the phase space position × wave vector satisfies a transport equation in the HF range. The transport equation allows to track the energy paths (rays) within the propagation medium, thus it also describes all power flows in slender structures. This theory provides a rational framework for the validation, and generalization, of heuristic methods such as the Statistical Energy Analysis (SEA, see [10]) or the Vibrational Conductivity Analogy (VCA, see [11]). Its main difficulty to date lies in the consideration of boundary and interface conditions for quadratic quantities, *e.g.* the energy and power flow densities, consistent with the boundary and interface conditions imposed to the displacement and stress fields of the original elastic wave system. The former are given as power flow reflection/transmission operators for the energy rays impinging on a boundary or an interface. Specular-like transverse boundary reflections, diffuse reflections, or fluid-structure coupling may be treated as a particular case [12]. A formal derivation of these operators at interfaces between slender sub-structures such as beams or shells has been proposed in [9] for the HF range. More recently, a systematic derivation of the elastic energy density boundary conditions near the doubly-hyperbolic set (where both compressional and shear elastic waves are incident on the boundary) and the hyperbolic-elliptic set (where shear waves are incident on the boundary with an angle greater than their critical angle) has been developed in [6]. The analysis however does not account for the glancing set (at critical incidence) and the transport of energy along an interface by so-called gliding rays; this issue is the subject of ongoing research. Yet a transport model in bounded media with general transverse or diffuse boundary conditions for the power flows may be derived from these results [13].

Transport equations can be solved numerically by various methods [14, 15], including nodal/spectral discontinuous "Galerkin" (DG) finite element methods [12, 16–20]. These techniques offer the great advantage of being numerically weakly dissipative and weakly dispersive–depending on the order of interpolation. Thus they are well suited for long-time simulations, when transport equations in bounded media converge to diffusion equations [21, 22]. The diffusive regime is characterized by energy equilibration rules which can be exhibited from direct numerical simulations using finite element or Monte-Carlo schemes [13]. Its onset at late times is prompted by the multiple scattering of energy rays on boundaries and interfaces. The diffusive regime is precisely the one invoked in the SEA and VCA approaches to derive steady-state global or local diffusion equation by heuristic arguments. The main purposes of this paper are twofold. First, the transport theory actually yielding diffusion for HF structural vibrations is outlined. Second, a discontinuous finite element method is developed to solve transient transport equations in view of demonstrating this asymptotic behavior at late times. It is organized as follows. First, the transport model for the evolution of the high-frequency energy density in heterogeneous structures is resumed, considering boundary and interface conditions as well. The DG method implemented for solving the corresponding transport equations is explained in section 3. Then two numerical examples are introduced in section 4: the first one deals with an assembly of thick beams, and the second one with a thick plate. At last section 5 offers some conclusions and directions of future works.

2 TRANSPORT MODEL

In section 2.1 one summarizes the basic results obtained in [3–6] for the transport properties of high-frequency waves in elastic media. The boundary conditions to be added to the evolution model recalled below are detailed in section 2.2 for specular-like or diffusive-like reflections at outer boundaries and in section 2.3 for reflections and transmissions at interfaces.

2.1 Transport in an open domain

Let us consider an heterogeneous, visco-elastic medium embedded in the open domain $\mathscr{O} \subseteq \mathbb{R}^d$, where $\mathbb{R}$ is the set of all real numbers and $d = 1, 2$ or 3 is the physical space dimension. Then the space-time energy density $\mathscr{E}(\mathbf{x},t) \in \mathbb{R}_+$ and power flow density $\Pi(\mathbf{x},t) \in \mathbb{R}^d$ associated to some strongly oscillating, high-frequency waves propagating in this medium may be written [3–6]:

$$\begin{aligned}\mathscr{E}(\mathbf{x},t) &= \frac{1}{2}\sum_{\alpha=1}^{M}\int_{\mathbb{R}^d} w_\alpha(\mathbf{x},\mathbf{k},t)\,\mathrm{d}\mathbf{k}\,,\\ \Pi(\mathbf{x},t) &= \frac{1}{2}\sum_{\alpha=1}^{M}\int_{\mathbb{R}^d} w_\alpha(\mathbf{x},\mathbf{k},t)\mathbf{c}_\alpha(\mathbf{x},\mathbf{k})\,\mathrm{d}\mathbf{k}\,,\end{aligned} \tag{1}$$

where M is the number of propagation modes, or polarizations in the mediun, $\{\mathbf{c}_\alpha\}_{1\leq\alpha\leq M}$ are their group velocities, and $\{w_\alpha\}_{1\leq\alpha\leq M}$ are angularly resolved energy densities for these modes living in the phase space $(\mathbf{x},\mathbf{k}) \in T^*\mathscr{O} \equiv \mathscr{O}\times\mathbb{R}^d$. Typically $M = 3$ and $\alpha =$ P, SV or SH in an isotropic elastic medium, for the compressional and shear waves respectively. These phase space densities w_α, also called hereafter specific intensities, satisfy in addition the multigroup transport equations (MTE):

$$\partial_t w_\alpha + \{\lambda_\alpha, w_\alpha\} = 0\,, \tag{2}$$

where $\lambda_\alpha(\mathbf{x},\mathbf{k}) = c_\alpha(\mathbf{x})|\mathbf{k}|$ is the eigenfrequency of the mode α, $\mathbf{c}_\alpha(\mathbf{x},\mathbf{k}) = \nabla_\mathbf{k}\lambda_\alpha(\mathbf{x},\mathbf{k}) = c_\alpha(\mathbf{x})\hat{\mathbf{k}}$, $\hat{\mathbf{k}} := \mathbf{k}/|\mathbf{k}|$ is the unit wave vector in $\mathbb{S}^{d-1}$, the unit sphere of $\mathbb{R}^d$, and $\{f,g\} = \nabla_\mathbf{k} f\cdot\nabla_\mathbf{x} g - \nabla_\mathbf{x} f\cdot\nabla_\mathbf{k} g$ is the usual Poisson bracket. It should be noted that, by the transport equations above, the specific intensities remain non-negative if the initial data are non-negative, as expected for energetic observables. These results are adapted to elastic waves in slender structures in [7–9], but they also apply to waves in poro-visco-elastic media, acoustic waves, electromagnetic waves, or the Schrödinger equation in random media [4].

For convenience, the MTE of equation (2) are reshaped in the form of a system of first-order conservative equations (3) in phase space $T^*\mathscr{O}$ as follows. Let us introduce the phase variable $\mathbf{z} = (\mathbf{x},\mathbf{k}) \in T^*\mathscr{O}$, the set $E = \{1,2,\dots M\}$, the vector $\mathbf{w} = (w_1, w_2, \dots w_M)^\mathsf{T}$ of $(\mathbb{R}_+)^M$, and the linear flux operator:

$$\mathscr{F}(\mathbf{w}) = (w_1\mathbf{F}_1, w_2\mathbf{F}_2, \dots w_M\mathbf{F}_M)^\mathsf{T}\,, \quad \mathbf{F}_\alpha(\mathbf{z}) = \begin{pmatrix}\mathbf{c}_\alpha(\mathbf{z})\\ -\nabla_\mathbf{x}\lambda_\alpha(\mathbf{z})\end{pmatrix}$$

($\mathscr{F}(\mathbf{w})$ is thus a $M\times 2d$ real matrix). Then equation (2) writes:

$$\partial_t\mathbf{w} + \mathbf{Div}_\mathbf{z}\mathscr{F}(\mathbf{w}) = \mathbf{0}\,, \tag{3}$$

which is a conservative system. On any discontinuity front Σ_D in phase space defined by the hypersurface $\Sigma_D = \{\mathbf{z}\in T^*\mathscr{O}; S(\mathbf{z}) = 0\}$, its piecewise continuous (weak) solution $\mathbf{w}$ satisfies the Rankine-Hugoniot condition:

$$\sum_{\alpha\in E}[\![\{\lambda_\alpha, S\}w_\alpha]\!] = 0\,, \tag{4}$$

where $[\![f]\!]$ stands for the jump of f on the front Σ_D. This relation states that the total normal power flow density $\sum_{\alpha\in E}\{\lambda_\alpha, S\}w_\alpha$ remains continuous accross that front; in return the specific intensities w_α have no reason to remain continuous themselves. The Rankine-Hugoniot condition (4) does not describe how the individual normal power flows $\{\lambda_\alpha, S\}w_\alpha = (\mathbf{F}_\alpha\cdot\hat{\mathbf{n}}_\Sigma)w_\alpha$ for each mode $\alpha\in E$ are distributed inward and outward on the front of unit outward normal $\hat{\mathbf{n}}_\Sigma = \hat{\nabla}_\mathbf{z} S$. This issue is considered in the next two sections for Σ_D being either a boundary or an interface (a junction) where the group velocities in the medium are discontinuous; in this case S is *a priori* independent of $\mathbf{k}$.

2.2 Transport in a bounded domain

The above system (3), completed by square-integrable initial conditions $\mathbf{w}(\mathbf{z},0) = \mathbf{w}^0(\mathbf{z})$, is subsequently considered in the phase space $T^*\mathscr{D} = \mathscr{D}\times\mathbb{R}^d$ where $\mathscr{D}$ is a bounded subset of $\mathscr{O}$, and for $t\in[0,T]$, $T < +\infty$ but arbitrary. It is assumed that $\mathbf{w}^0(\mathbf{x},\{\mathbf{k} = \mathbf{0}\}) = \mathbf{0}$. Wavenumber $|\mathbf{k}| > 0$ is a fixed parameter, and we shall also assume as usual that the velocities $c_\alpha \in W^{1,\infty}(\mathscr{D})$ are positive functions with possible isolated discontinuities if $\mathscr{D}$ contains sharp interfaces between different materials. For any smooth manifold Σ of codimension 1 in $\mathbb{R}^d$ oriented by its normal $\hat{\mathbf{n}}_\Sigma$, let:

$$\begin{aligned}\Gamma^\pm(\Sigma) &= \{(\mathbf{x},\mathbf{k})\in\Sigma\times\mathbb{R}^d; \pm\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}_\Sigma(\mathbf{x}) > 0\}\,,\\ \Gamma^\pm_\mathbf{x}(\Sigma) &= \{\mathbf{k}\in\mathbb{R}^d; \pm\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}_\Sigma(\mathbf{x}) > 0\}\,.\end{aligned}$$

Then well-posedness of the problem is ensured by setting proper boundary conditions on the inflow boundary $\Gamma^-(\partial\mathscr{D})$ of the computational domain $\mathscr{D}$ [23]. Denoting by $\hat{\mathbf{n}}$ the outward unit normal to the (smooth) boundary $\partial\mathscr{D}$, an incoming flux $\mathbf{f}(\mathbf{z},t)$ can be imposed as $-\mathscr{F}(\mathbf{w})\hat{\mathbf{n}} = \mathbf{f}$ on $\Gamma^-(\partial\mathscr{D})$ and for $t > 0$, where $\mathscr{F}(\mathbf{w})\hat{\mathbf{n}} := ((\mathbf{F}_1\cdot\hat{\mathbf{n}})w_1, (\mathbf{F}_2\cdot\hat{\mathbf{n}})w_2, \dots(\mathbf{F}_M\cdot\hat{\mathbf{n}})w_M)^\mathsf{T}$. However generalized boundary reflection laws can also be considered in the form:

$$\mathscr{R}_\mathbf{x}(\mathbf{w}) + \mathbf{f} = \mathbf{0} \quad \text{on}\ \Gamma^-_\mathbf{x}(\partial\mathscr{D})\,, \quad t > 0\,, \tag{5}$$

where $\mathscr{R}_\mathbf{x}$ is a so-called boundary reflection operator, which depends only on the local geometry of the boundary (it may also depend on $|\mathbf{k}|$). This operator describes how the outward normal power flows $\mathscr{F}(\mathbf{w})\hat{\mathbf{n}}$, $\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}>0$, are reflected into inward normal power flows $-\mathscr{F}(\mathbf{w})\hat{\mathbf{n}}$, $\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}<0$, with possible mode conversions in E. Here the glancing set $\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}=0$ is ignored, not because such grazing rays are unlikely to occur, but because we lack a theoretical model to describe the associated boundary specific intensities at present. The rigourous construction of $\mathscr{R}_{\mathbf{x}}$ away from the glancing set is detailed in [6] for elastic waves, while a formal derivation for bending, compressional and shear waves in thick beams and shells is given in [9]. The α-th component of this operator has the general form:

$$\mathscr{R}_{\mathbf{x}}(\mathbf{w})_\alpha = \sum_{\beta=1}^{M}\int_{\hat{\mathbf{p}}\cdot\hat{\mathbf{n}}>0}\rho_{\alpha\beta}(\mathbf{x},\hat{\mathbf{k}},\mathrm{d}\hat{\mathbf{p}}) \times\left(|\mathbf{F}_\beta\cdot\hat{\mathbf{n}}|w_\beta(\mathbf{x},\mathbf{p})-|\mathbf{F}_\alpha\cdot\hat{\mathbf{n}}|w_\alpha(\mathbf{x},\mathbf{k})\right)\quad\text{on }\Gamma^-_{\mathbf{x}}(\partial\mathscr{D}),\qquad(6)$$

for a conservative boundary such that:

$$\sum_{\beta=1}^{M}\int_{\hat{\mathbf{p}}\cdot\hat{\mathbf{n}}<0}\rho_{\beta\alpha}(\mathbf{x},\mathrm{d}\hat{\mathbf{p}},\hat{\mathbf{k}})=1\quad\text{on }\Gamma^+_{\mathbf{x}}(\partial\mathscr{D}),$$

with $\rho_{\beta\alpha}(\mathbf{x},\hat{\mathbf{p}},\hat{\mathbf{k}})=\rho_{\alpha\beta}(\mathbf{x},-\hat{\mathbf{k}},-\hat{\mathbf{p}})=\rho_{\alpha\beta}(\mathbf{x},\hat{\mathbf{k}}^\star,\hat{\mathbf{p}}^\star)$, where $\hat{\mathbf{k}}^\star(\hat{\mathbf{k}}):=(\mathbb{I}_d-2\hat{\mathbf{n}}\otimes\hat{\mathbf{n}})\hat{\mathbf{k}}$. Specular reflections for example are $\rho_{\alpha\beta}(\mathbf{x},\hat{\mathbf{k}},\mathrm{d}\hat{\mathbf{p}})=\delta_{\alpha\beta}\otimes\delta(\hat{\mathbf{p}}-\hat{\mathbf{k}}^\star)$, $\delta_{\alpha\beta}$ standing for the Kronecker symbol; see [13]. These kernels are the power reflection coefficients of the boundary, or reflectances in the optics literature.

2.3 Transport with a sharp interface

It is now considered that the physical domain $\mathscr{D}$ is partitioned into non-overlapping sub-domains and their interface $\Sigma_D\subset\overline{\mathscr{D}}$ is a smooth, bounded manifold of codimension 1 oriented by its unit normal $\hat{\mathbf{n}}_\Sigma$ (ignoring edges and corners at a first glance). More specifically, Σ_D is any interface between these sub-domains or between a sub-domain and the exterior of $\mathscr{D}$ and consequently, it may also be a part or the whole of $\partial\mathscr{D}$. If the group velocities c_α are continuous across Σ_D, the continuity of the normal power flow on Σ_D is ensured by the conditions $\mathscr{F}(\mathbf{w}^+)\hat{\mathbf{n}}_\Sigma=\mathscr{F}(\mathbf{w}^-)\hat{\mathbf{n}}_\Sigma$ if $\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}_\Sigma>0$ (forward flux) and $-\mathscr{F}(\mathbf{w}^-)\hat{\mathbf{n}}_\Sigma=-\mathscr{F}(\mathbf{w}^+)\hat{\mathbf{n}}_\Sigma$ if $\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}_\Sigma<0$ (backward flux), where $\mathbf{w}^\pm(\mathbf{x},\mathbf{k},t)=\lim_{h\downarrow 0}\mathbf{w}(\mathbf{x}\pm h\hat{\mathbf{n}}_\Sigma,\mathbf{k},t)$ for a.e. $\mathbf{x}\in\Sigma_D$, and $\mathbf{w}^+(\mathbf{x},\mathbf{k},t)=\mathbf{0}$ whenever $\mathbf{x}\in\partial\mathscr{D}$; likewise, $c^\pm_\alpha(\mathbf{x})=\lim_{h\downarrow 0}c_\alpha(\mathbf{x}\pm h\hat{\mathbf{n}}_\Sigma)$ a.e. $\mathbf{x}\in\Sigma_D$. If however the group velocities are discontinuous across Σ_D, such that $c^+_\alpha\neq c^-_\alpha$, these expressions may be generalized to the form:

$$\begin{aligned}\mathscr{R}^-_{\mathbf{x}}(\mathbf{w}^-)+\mathscr{T}^+_{\mathbf{x}}(\mathbf{w}^-,\mathbf{w}^+)&=\mathbf{0}\quad\text{on }\Gamma^-_{\mathbf{x}}(\Sigma_D),\\ \mathscr{R}^+_{\mathbf{x}}(\mathbf{w}^+)+\mathscr{T}^-_{\mathbf{x}}(\mathbf{w}^+,\mathbf{w}^-)&=\mathbf{0}\quad\text{on }\Gamma^+_{\mathbf{x}}(\Sigma_D),\end{aligned}\qquad(7)$$

$t>0$, where $\mathscr{R}^\pm_{\mathbf{x}}$ and $\mathscr{T}^\mp_{\mathbf{x}}$ are the upstream $(-)$ and downstream $(+)$ reflection and transmission operators, respectively, of the interface Σ_D. The former have expressions similar to equation (6) on either side of Σ_D, while the latter have the form:

$$\mathscr{T}^+_{\mathbf{x}}(\mathbf{w}^-,\mathbf{w}^+)_\alpha=\sum_{\beta=1}^{M}\int_{\hat{\mathbf{p}}\cdot\hat{\mathbf{n}}_\Sigma<0}\tau^+_{\alpha\beta}(\mathbf{x},\hat{\mathbf{k}},\mathrm{d}\hat{\mathbf{p}}) \times\left(|\mathbf{F}_\beta\cdot\hat{\mathbf{n}}_\Sigma|w^+_\beta(\mathbf{x},\mathbf{p})-|\mathbf{F}_\alpha\cdot\hat{\mathbf{n}}_\Sigma|w^-_\alpha(\mathbf{x},\mathbf{k})\right)\quad\text{on }\Gamma^-_{\mathbf{x}}(\Sigma_D),\qquad(8)$$

interchanging the $+$ and $-$ signs for $\mathscr{T}^-_{\mathbf{x}}(\mathbf{w}^+,\mathbf{w}^-)$. Their kernels–the power transmission coefficients of the interface or transmittances–satisfy the symmetries $\tau^-_{\beta\alpha}(\mathbf{x},\hat{\mathbf{p}},\hat{\mathbf{k}})=\tau^+_{\alpha\beta}(\mathbf{x},-\hat{\mathbf{k}},-\hat{\mathbf{p}})=\tau^+_{\alpha\beta}(\mathbf{x},\hat{\mathbf{k}}^\star,\hat{\mathbf{p}}^\star)$ for a conservative interface such that:

$$\sum_{\beta=1}^{M}\left(\int_{\pm\hat{\mathbf{p}}\cdot\hat{\mathbf{n}}_\Sigma>0}\rho^\pm_{\beta\alpha}(\mathbf{x},\mathrm{d}\hat{\mathbf{p}},\hat{\mathbf{k}})+\int_{\pm\hat{\mathbf{p}}\cdot\hat{\mathbf{n}}_\Sigma<0}\tau^\pm_{\beta\alpha}(\mathbf{x},\mathrm{d}\hat{\mathbf{p}},\hat{\mathbf{k}})\right)=1$$

on $\Gamma^\mp_{\mathbf{x}}(\Sigma_D)$. The relations (7) show how the inward and outward normal power flows are distributed on the interface: the outward flows are the sum of reflected and transmitted flows, taking into account possible mode conversions in E, on either side of that interface. The explicit derivation of the reflection and transmission operators for junctions of thick beams or shells is outlined in [9]. Their general properties and some examples are also given in [13].

3 NUMERICAL METHOD: DISCONTINUOUS FINITE ELEMENTS

Numerical methods with low numerical dispersion and dissipation errors are needed to perform long-time simulations of the MTE in order to reach its diffusion limit. Monte-Carlo methods [24] and discontinuous finite element methods [12, 16–20] both have these properties. Monte-Carlo methods are very easy to implement since they are based on an intuitive interpretation of transport equations. They also allow to compute local solutions, a decisive advantage over energetic methods (such as finite elements) for large-scale computations. Their main drawback is their lack of versatility for complex geometries as typically encountered in structural dynamics and acoustics; they will not be considered in the following, but the basic principles and some numerical experiments can be found in [25, 26]. Finite element methods are, on the contrary, much more flexible and can be applied to truly complex geometries. Among these methods, the discontinuous "Galerkin" (DG) finite element method has originally been introduced by Reed & Hill [16] and Lesaint & Raviart [17] to compute neutronic transfers; it is therefore adapted to the integration of MTEs. In the DG method, boundary fluxes at the edges or faces of the elements maintain the consistency of the numerical solution with the continuous, non-discretized MTEs, which are otherwise discretized on each element without any continuity relation between elements; this scheme is recalled in section 3.1 below. The numerical fluxes are constructed in order to get to satisfy the boundary and interface conditions (5) and (7); some implementation issues are discussed in section 3.2.

3.1 Basic principles of the DG method

Let us consider the subdivision $\mathscr{T}_h=\cup_{r=1}^{\mathscr{N}}\mathsf{D}_r$ of $\mathscr{D}$ into $\mathscr{N}$ non-overlapping sub-domains, or finite elements D_r. A Galerkin variational formulation of the system (3) in an *ad hoc* functional set $\mathscr{W}_h$ defined on $\mathscr{T}_h$ is: Find $\mathbf{w}\in\mathscr{W}_h$ such that

$$\int_{\mathsf{D}_r\times\mathbb{R}^d}\left(\partial_t\mathbf{w}\cdot\mathbf{v}-\mathscr{F}(\mathbf{w}):\nabla_{\mathbf{z}}\mathbf{v}^{\mathsf{T}}\right)\mathrm{d}\mathbf{z} +\int_{\partial\mathsf{D}_r\times\mathbb{R}^d}\mathscr{F}(\mathbf{w})\hat{\mathbf{n}}_r\cdot\mathbf{v}\,\mathrm{d}\gamma(\mathbf{z})=0,\quad\forall\mathbf{v}\in\mathscr{W}_h,\qquad(9)$$

where $\hat{\mathbf{n}}_r$ is outward unit normal to D_r. If the set $\mathscr{W}_h$ is continuous over adjacent subdomains, summing the above expression yields the standard continuous variational formulation:

$$\int_{T^*\mathscr{D}} \left(\partial_t \mathbf{w}\cdot\mathbf{v} - \mathscr{F}(\mathbf{w}) : \nabla_{\mathbf{z}}\mathbf{v}^{\mathsf{T}}\right) \mathrm{d}\mathbf{z} + \int_{T^*\partial\mathscr{D}} \mathscr{F}(\mathbf{w})\hat{\mathbf{n}}\cdot\mathbf{v}\,\mathrm{d}\gamma(\mathbf{z}) = 0\,, \quad \forall \mathbf{v}\in\mathscr{W}_h\,. \quad (10)$$

If however $\mathscr{W}_h$ is discontinous over two adjacent sub-domains, then the normal flux $\mathscr{F}(\mathbf{w})\hat{\mathbf{n}}_r$ in equation (9) has to be replaced by a numerical flux $\mathscr{F}_r^*(\mathbf{w}^-,\mathbf{w}^+)\hat{\mathbf{n}}_r$ because the left and right traces of $\mathbf{w}$ on $\partial\mathsf{D}_r$, denoted by $\mathbf{w}^-$ and $\mathbf{w}^+$ respectively, are different *a priori*. The corresponding discontinuous variational formulation is:

$$\int_{\mathsf{D}_r\times\mathbb{R}^d} \left(\partial_t \mathbf{w}\cdot\mathbf{v} - \mathscr{F}(\mathbf{w}) : \nabla_{\mathbf{z}}\mathbf{v}^{\mathsf{T}}\right) \mathrm{d}\mathbf{z} + \int_{\partial\mathsf{D}_r\times\mathbb{R}^d} \mathscr{F}_r^*(\mathbf{w}^-,\mathbf{w}^+)\hat{\mathbf{n}}_r\cdot\mathbf{v}\,\mathrm{d}\gamma(\mathbf{z}) = 0\,, \quad \forall \mathbf{v}\in\mathscr{W}_h\,, \quad (11)$$

with $\mathbf{w}^\pm = \lim_{h\downarrow 0}\mathbf{w}(\mathbf{x}\pm h\hat{\mathbf{n}}_r,\mathbf{k},t)$ a.e. $\mathbf{x}\in\partial\mathsf{D}_r$, and the convention:

$$\mathscr{F}_r^*(\mathbf{w}^-,\mathbf{w}^+)\hat{\mathbf{n}} = \mathscr{F}(\mathbf{w}^-)\hat{\mathbf{n}}\,, \quad \mathbf{w}^+ = \mathbf{0}\,, \quad \mathbf{z}\in\Gamma^-(\partial\mathscr{D})\,.$$

The numerical flux has also to be consistent, that is $\mathscr{F}_r^*(\mathbf{w},\mathbf{w}) = \mathscr{F}(\mathbf{w})$, and conservative, that is $\mathscr{F}_r^*(\mathbf{w}^-,\mathbf{w}^+)\hat{\mathbf{n}}_r + \mathscr{F}_s^*(\mathbf{w}^-,\mathbf{w}^+)\hat{\mathbf{n}}_s = \mathbf{0}$ if $\partial\mathsf{D}_r\cap\partial\mathsf{D}_s \neq 0$. The aforementioned formulation is the so-called "weak" form of the DG method [20]. The "strong" form corresponding to the original DG method introduced by Lesaint & Raviart [17] is derived by integrating the weak form by parts. Let $\mathscr{L}\mathbf{w} := \partial_t\mathbf{w} + \nabla_{\mathbf{z}}\cdot\mathscr{F}(\mathbf{w})$, then:

$$\int_{\mathsf{D}_r\times\mathbb{R}^d} \mathscr{L}\mathbf{w}\cdot\mathbf{v}\,\mathrm{d}\mathbf{z} = \int_{\partial\mathsf{D}_r\times\mathbb{R}^d} \left(\mathscr{F}(\mathbf{w}^-) - \mathscr{F}_r^*(\mathbf{w}^-,\mathbf{w}^+)\right)\hat{\mathbf{n}}_r\cdot\mathbf{v}\,\mathrm{d}\gamma(\mathbf{z})\,, \quad \forall \mathbf{v}\in\mathscr{W}_h\,. \quad (12)$$

This last form can also be seen as a penalization method applied to the boundary fluxes of the sub-domains (or elements). An alternative form for linear systems is the ultra-weak variational formulation proposed by Després [27]. It consists in choosing $\mathscr{W}_h = \{\mathbf{v}\,,\ \mathscr{L}^*\mathbf{v} = \mathbf{0}\}$, where $\mathscr{L}^*\mathbf{w} \equiv -\partial_t\mathbf{w} - (\mathbf{F}_\alpha\cdot\nabla_{\mathbf{z}}w_\alpha)_{\alpha\in E}$ is the formal adjoint of $\mathscr{L}$. Then (12) is written:

$$\partial_t\left(\int_{\mathsf{D}_r\times\mathbb{R}^d} \mathbf{w}\cdot\mathbf{v}\,\mathrm{d}\mathbf{z}\right) + \int_{\partial\mathsf{D}_r\times\mathbb{R}^d} \mathscr{F}_r^*(\mathbf{w}^-,\mathbf{w}^+)\hat{\mathbf{n}}_r\cdot\mathbf{v}\,\mathrm{d}\gamma(\mathbf{z}) = 0\,, \quad \forall \mathbf{v}\in\mathscr{W}_h\,. \quad (13)$$

The test functions $\mathbf{v}$ have then to be taken as solutions of an adjoint problem, typically plane waves in piecewise homogeneous media [28]: $\mathbf{v}(\mathbf{z},t) = \mathbf{V}\phi(\mathbf{k}\cdot\mathbf{x} - \omega t)$ such that $\Gamma(\mathbf{z})\mathbf{V} = \omega\mathbf{V}$, meaning that ω and $\mathbf{V}$ are the eigenvalues and eigenvectors of the acoustic tensor Γ which is simply $\Gamma(\mathbf{z}) = \mathrm{diag}(\lambda_\alpha(\mathbf{z}))_{\alpha\in E}$ here.

3.2 Numerical implementation

The numerical implementation of the DG method expounded above requires three main ingredients, once the subdivision $\mathscr{T}_h$ has been performed: (i) the definition of numerical fluxes $\mathscr{F}_r^*(\mathbf{w}^-,\mathbf{w}^+)$ on all elements, (ii) the choice of the approximation set $\mathscr{W}_h$, and (iii) the choice of a time-integration scheme. The computational domain in phase space is reduced to the unit cosphere bundle $S^*\mathscr{D} = \mathscr{D}\times\mathbb{S}^{d-1}$ by considering only normalized wave vectors $\hat{\mathbf{k}}$ defined on the unit sphere $\mathbb{S}^{d-1}$ of $\mathbb{R}^d$. For applications to assemblies of slender structures, $d=1$ (beams) and $\mathbb{S}^0 = \{-1,+1\}$, or $d=2$ (shells) and the unit circle $\mathbb{S}^1$ is parametrized by $\theta\in[0,2\pi[$ such that $\hat{\mathbf{k}} = (\cos\theta,\sin\theta)^{\mathsf{T}}$. In both cases the meshing of $S^*\mathscr{D}$ writes $\mathscr{T}_h = \cup_{r=1}^{\mathscr{N}}\mathsf{X}_r = \cup_{r=1}^{\mathscr{N}}\mathsf{D}_r\times\mathbb{S}^{d-1}$, denoting by h the largest diameter of the elements D_r.

Regarding (i), the numerical fluxes ensuring the stability and consistency of the DG method for transport equations have the form [19]:

$$\mathscr{F}_r^*(\mathbf{w}^-,\mathbf{w}^+) = \frac{1}{2}\left(\mathscr{F}(\mathbf{w}^-) + \mathscr{F}(\mathbf{w}^+)\right) + \mathbf{C}_r\left(\mathbf{w}^- - \mathbf{w}^+\right)\hat{\mathbf{n}}_r$$

where $\mathbf{C}_r$ is an $M\times M$ non-negative definite matrix. For example, the (upwind) Roe flux corresponds to $\mathbf{C}_r = \frac{1}{2}\mathrm{diag}(|\mathbf{F}_\alpha\cdot\hat{\mathbf{n}}_r|)_{\alpha\in E}$ and the Lax-Friedrichs flux corresponds to $\mathbf{C}_r = \frac{1}{2}\mathrm{diag}(|\mathbf{F}_\alpha|)_{\alpha\in E}$. This expression can be generalized as follows for an element boundary $\partial\mathsf{D}_r$ belonging to an interface (a junction) Σ_D where the celerities c_α are discontinuous. Let $\Sigma_D^r = \partial\mathsf{D}_r\cap\Sigma_D \neq \emptyset$, then having in mind equation (7), the (upwind) numerical flux is written:

$$\mathscr{F}_r^*(\mathbf{w}^-,\mathbf{w}^+)\hat{\mathbf{n}}_r = \mathscr{F}(\mathbf{w}^-)\hat{\mathbf{n}}_r \quad (14)$$

on $\Gamma_{\mathbf{x}}^+(\Sigma_D^r)$, and:

$$\mathscr{F}_r^*(\mathbf{w}^-,\mathbf{w}^+)\hat{\mathbf{n}}_r = \mathscr{F}(\mathbf{w}^-)\hat{\mathbf{n}}_r - \left(\mathscr{R}_{\mathbf{x}}^-(\mathbf{w}^-) + \mathscr{T}_{\mathbf{x}}^+(\mathbf{w}^-,\mathbf{w}^+)\right) \quad (15)$$

on $\Gamma_{\mathbf{x}}^-(\Sigma_D^r)$. This definition extends in fact to the case $\Sigma_D^r = \emptyset$ if the celerities are continuous across $\partial\mathsf{D}_r$. Then $\mathscr{R}_{\mathbf{x}}^-(\mathbf{w}^-) \equiv \mathbf{0}$ and $\mathscr{T}_{\mathbf{x}}^+(\mathbf{w}^-,\mathbf{w}^+) = (\mathscr{F}(\mathbf{w}^-) - \mathscr{F}(\mathbf{w}^+))\hat{\mathbf{n}}_r$ on $\Gamma_{\mathbf{x}}^-(\partial\mathsf{D}_r)$, such that $\mathscr{F}_r^*(\mathbf{w}^-,\mathbf{w}^+) = \mathscr{F}(\mathbf{w}^-)$ on $\Gamma_{\mathbf{x}}^+(\partial\mathsf{D}_r)$ and $\mathscr{F}_r^*(\mathbf{w}^-,\mathbf{w}^+) = \mathscr{F}(\mathbf{w}^+)$ on $\Gamma_{\mathbf{x}}^-(\partial\mathsf{D}_r)$.

Regarding (ii), the approximation set $\mathscr{W}_h \subset L^2(S^*\mathscr{D})$ is constructed by tensoring the physical space and phase variables $\mathscr{W}_h = \mathscr{D}_h\otimes\mathscr{S}_h$, with:

$$\mathscr{D}_h = \{\mathbf{v}\in(L^2(\mathscr{D}))^M;\, v_\alpha|_{\mathsf{D}_r}\in\mathbb{P}_p(\mathsf{D}_r),\, 1\leq r\leq\mathscr{N},\, \alpha\in E\}$$

where $\mathbb{P}_p(\mathsf{D}_r)\subset H^1(\mathsf{D}_r)$ is typically a polynomial set, and $\mathscr{S}_h$ is a finite dimensional set of real-valued functions defined on $\mathbb{S}^{d-1}$. Practically in dimension 1, $\mathbb{P}_p(\mathsf{D}_r)$ is spanned by either a set of Jacobi orthogonal polynomials up to the order p (the modal, or spectral approach), or a set of Lagrange interpolation polynomials associated to $p+1$ interpolation points (the nodal approach). The degrees of freedom in physical space are the projection coefficients on the Jacobi polynomials of the approximate solution in $\mathscr{W}_h$ in the modal approach, and its values at the interpolation points in the nodal approach. It should be noted that if the latter are chosen as Gauss-Jacobi quadrature points on $[-1,1]$, the corresponding Lagrange polynomials are computed explicitly as functions of Jacobi polynomials, which yields a unified framework for the modal and nodal approaches. The numerical dispersion and dissipation errors introduced by this spatial discretization are $O(K^{2p+3})$ and $O(K^{2p+2})$, respectively, for decentered numerical fluxes ($\mathbf{C}_r \neq \mathbf{0}$), or $O(K^{4\lfloor \frac{p}{2} \rfloor+3})$ and 0 for centered fluxes ($\mathbf{C}_r = \mathbf{0}$) [29], where $K = h|\mathbf{k}|$ and $\lfloor\cdot\rfloor$ stands for the integer part. These so-called "superconvergence" properties of DG methods are particularly attractive for long-times numerical simulations. The spatial discretization in dimension 2 is achieved by tensoring both dimensions on structured meshes, as proposed in [30] for example. The construction of $\mathscr{S}_h$ is however more involved. The modal approach corresponds to a truncated Fourier expansion in θ, but this scheme does not ensure the positivity of the approximate solution because of the slowness of the convergence of Fourier series. The nodal approach by Lagrange interpolation considered in [31] yields the well-known ray effect, which also arises in the discrete ordinate methods widely used in neutronic or thermal transfer computations [14, chap. 8]. Therefore further researches are needed on this issue; it is the subject of ongoing investigations.

At last, regarding (iii), strong stability-preserving (SSP), high-order Runge-Kutta schemes have been implemented [32]. The CFL condition of these schemes, coupled to the aforementioned phase space discretization methods, can be computed numerically for various orders p.

4 NUMERICAL EXAMPLES

In the examples below the transport equations (2) with specular-like boundary and interface (at the junctions) conditions are solved numerically by the schemes expounded in the foregoing section. In section 4.1 the example of [9] of scalar transport in coupled Timoshenko beams is resumed and extended, whereas section 4.2 deals with a thick plate. In both cases the weak form (11) of the DG method is used for the simulations.

4.1 Coupled beams

We first consider an assembly of two Timoshenko beams $\mathscr{D} =]-L,0[\cup]0,L[$, $L>0$, with a junction at $x=0$ such that beam #2 forms an angle ϕ with beam #1. Three types of energy rays propagate in such a system: a pure shear transverse mode at the velocities $c_{\mathrm{T}j}$ ($j = 1$ or 2 for either beam #1 or beam #2) denoted by subscript T, a pure bending mode at the velocities $c_{\mathrm{P}j}$ denoted by subscript Pb, and a longitudinal mode at the velocities $c_{\mathrm{P}j}$ denoted by subscript Pn; see [8,9] for some details on the derivation. Therefore the energy density evolution in this system is depicted by three coupled, one-dimensional transport equations ($M=3$ and $d=1$ such that $\hat{k} \equiv \mathrm{sign(k)}$) for $\alpha = \mathrm{T},\mathrm{Pb}$ or Pn written as:

$$\partial_t w_\alpha + c_\alpha \hat{k} \partial_x w_\alpha = 0\,, \tag{16}$$

where the coupling is ensured by the left ($-$) and right ($+$) power reflection and transmission operators $\mathscr{R}_0^\pm$, $\mathscr{T}_0^\pm$ of the junction. The expressions of their kernels $\rho^\pm_{\alpha\beta}(0,\hat{k},\hat{p})$ and $\tau^\pm_{\alpha\beta}(0,\hat{k},\hat{p})$ as functions of ϕ and beams mechanical and geometrical characteristics are derived in [9], using a high-frequency model compatible with the propagation characteristics obtained by the transport theory. This study has also shown that the power reflection coefficients at a beam end for either Dirichlet (clamped) or Neumann (free) boundary condition were all equal to 1, without mode conversion of any type; therefore specular reflections are considered at the boundaries $x = \pm L$. All computations are performed for the initial conditions:

$$w^0_{\mathrm{T}}(x,k) = \varphi\left(\frac{x-x_0}{R_0}\right) \otimes \delta(\hat{k}-\hat{k}_0)\,, \quad w^0_{\mathrm{Pb}} = w^0_{\mathrm{Pn}} \equiv 0\,,$$

with $\varphi(x) = \max(0, 1-2|x|)$ ("hat" source) or $\varphi(x) = \mathbf{1}_{\{|x|\leq\frac{1}{2}\}}(x)$ ("square" source). The parameters are chosen as $x_0 = -0.1 \times L$, $R_0 = 0.12 \times L$. Beam velocities are such that $c_{\mathrm{T}2}/c_{\mathrm{T}1} = c_{\mathrm{P}2}/c_{\mathrm{P}1} = \sqrt{2}$, their Poisson's ratio being $\nu_1 = \nu_2 = 0.3$, and the junction angle is $\phi = \pi/4$.

The coupled transport equations (16) are solved numerically by the Runge-Kutta DG finite element method summarized in section 3.2 above. The assembly is partitioned into 40 uniform spatial elements, and equations (16) are solved in time by a eight-order SSP Runge-Kutta scheme. The Courant number for the simulations is fixed at $\mathrm{CFL} \equiv c_{\mathrm{T}2}\frac{\Delta t}{\Delta x} = 2.5\,10^{-2}$. Figure 1 and figure 2 display our results for either the "hat" source with $\hat{k}_0 = +1$, or the "square" source with $\hat{k}_0 = -1$; tenth-order Lagrange polynomial bases, using Legendre Gauss-Lobatto interpolation points, are used in both cases. As for the "square" source, the reconstructed numerical solution has been filtered in order to tame the Gibbs phenomenon [33]. The time scale is $T = L/c_{\mathrm{T}2}$, the time needed by the wave front to reach the extremity $x = L$ of the homogeneous assembly starting from the junction. The thin vertical line indicates the location of the junction. The wavefronts of shear ($\alpha = \mathrm{T}$) and bending/axial ($\alpha = \mathrm{P}$) energy densities propagating at the speeds $c_{\mathrm{T}j}$ and $c_{\mathrm{P}j}$, respectively, are marked by small capital letters on figure 1. Figure 3 and figure 4 display the overall mechanical energies $\mathscr{E}^h_j(t)$ computed from the numerical solutions $w^h_\alpha(x,\hat{k},t)$ of (16) within each beam by:

$$\mathscr{E}^h_1(t) = \sum_{\alpha\in E}\int_{-L}^0 w^h_\alpha(x,\pm1,t)\mathrm{d}x\,,$$

$$\mathscr{E}^h_2(t) = \sum_{\alpha\in E}\int_0^L w^h_\alpha(x,\pm1,t)\mathrm{d}x\,,$$

and their sum, normalized by the total energy introduced into the system $\mathscr{E}(0) = \sum_\alpha \mathscr{E}^0_\alpha = \mathscr{E}^0_{\mathrm{T}}$. This normalized sum is equal to 1 for all times: one concludes that the proposed scheme in dimension one is numerically conservative.

These results exhibit the following features: one observes a beating phenomenon of the total energy within each subsystem, by which it is transferred back and forth from the first beam to the second one and the other way round. At late time an equipartition regime holds between both beams. It is described by diffusion equations [21, 22]. The onset of diffusion is prompted by the mixing of wave fronts and modes at the junction, and to a lesser extent at the boundaries. This mixing process may be accelerated by the presence of random heterogeneities, as demonstrated in [13]. Finally, the long-time limit of the ratio $\mathscr{E}^h_{\mathrm{P}j}(t)/\mathscr{E}^h_{\mathrm{T}j}(t)$ of the bending/longitudinal energy over the shear energy within each beam is expected to be the ratio $c_{\mathrm{T}j}/c_{\mathrm{P}j}$ to the power of the physical dimension ($d=1$ in the present case) independently of the source type, in accordance with the results of [4, 34] for example. Here this universal equilibration rule is recovered by the DG scheme, as examplified by figure 5 and figure 6 for both sources.

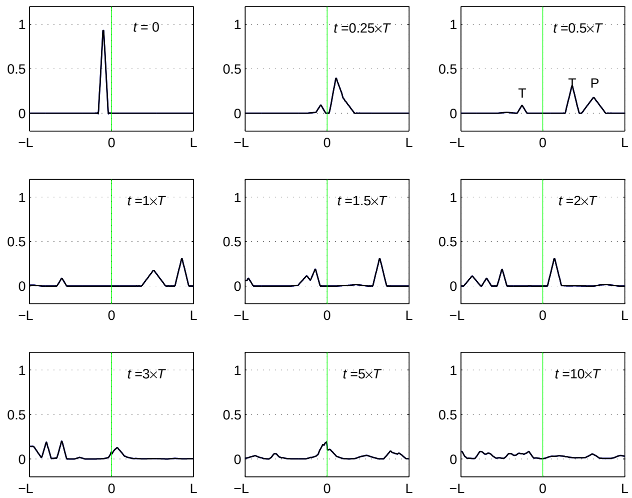

Figure 1. Evolution of the energy density in an assembly of two Timoshenko beams for $\phi = \frac{\pi}{4}$, $c_{\mathrm{T2}}/c_{\mathrm{T1}} = c_{\mathrm{P2}}/c_{\mathrm{P1}} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $T = L/c_{\mathrm{T2}}$, and a "hat" source with $\hat{k}_0 = +1$. 8-th order SSP Runge-Kutta DG method with 10-th order Legendre Gauss-Lobatto nodal expansion and $\mathscr{N} = 40$ elements.

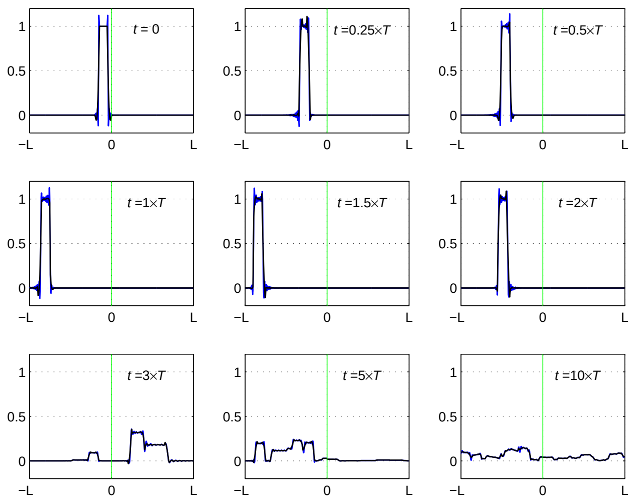

Figure 2. Evolution of the energy density in an assembly of two Timoshenko beams for $\phi = \frac{\pi}{4}$, $c_{\mathrm{T2}}/c_{\mathrm{T1}} = c_{\mathrm{P2}}/c_{\mathrm{P1}} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $T = L/c_{\mathrm{T2}}$, and a "square" source with $\hat{k}_0 = -1$. 8-th order SSP Runge-Kutta DG method with 10-th order Legendre Gauss-Lobatto nodal expansion and $\mathscr{N} = 40$ elements, — unfiltered solution, — exponential filter.

4.2 *A flat plate*

Next, a flat plate $\mathscr{D} =]-L,L[\times]-L,L[$ with free boundary conditions is considered. Five types of energy rays propagate in this structure: a pure shear mode at the velocity c_{T} denoted by subscript T, two longitudinal and transverse bending modes at the velocities c_{P} and c_{S} denoted by subscripts Pb and

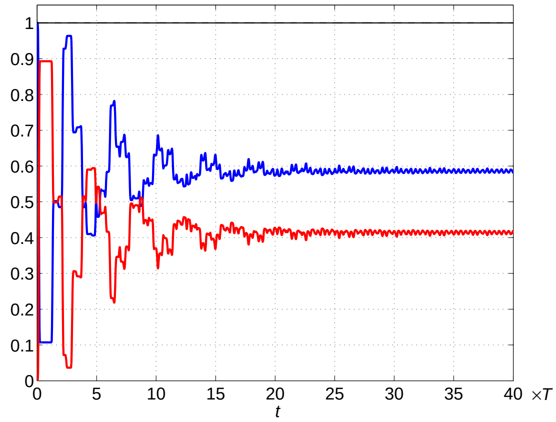

Figure 3. Evolution of the total vibrational energies in an assembly of two Timoshenko beams for $\phi = \frac{\pi}{4}$, $c_{\mathrm{T2}}/c_{\mathrm{T1}} = c_{\mathrm{P2}}/c_{\mathrm{P1}} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $T = L/c_{\mathrm{T2}}$, and a "hat" source with $\hat{k}_0 = +1$. 8-th order SSP Runge-Kutta DG method with 10-th order Legendre Gauss-Lobatto nodal expansion and $\mathscr{N} = 40$ elements, — beam #1, — beam #2, — computed total energy.

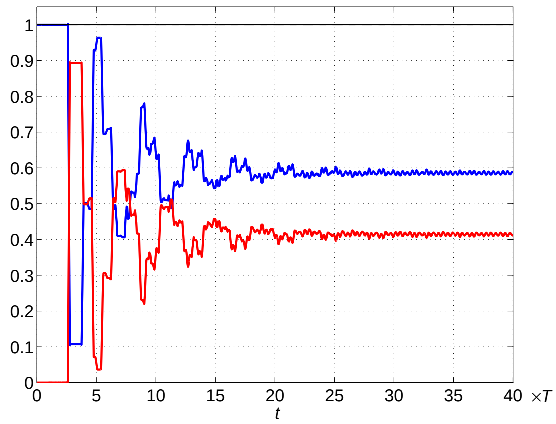

Figure 4. Evolution of the total vibrational energies in an assembly of two Timoshenko beams for $\phi = \frac{\pi}{4}$, $c_{\mathrm{T2}}/c_{\mathrm{T1}} = c_{\mathrm{P2}}/c_{\mathrm{P1}} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $T = L/c_{\mathrm{T2}}$, and a "square" source with $\hat{k}_0 = -1$. 8-th order SSP Runge-Kutta DG method with 10-th order Legendre Gauss-Lobatto nodal expansion and $\mathscr{N} = 40$ elements, — beam #1, — beam #2, — computed total energy.

Sb, respectively, and two longitudinal and transverse in-plane modes at the same velocities denoted by subscript Pn and Sn; see [7,9]. Therefore the energy density evolution in the plate is depicted by five coupled, two-dimensional transport equations ($M = 5$ and $d = 2$) for $\alpha \in E = \{\mathrm{T},\mathrm{Pb},\mathrm{Pn},\mathrm{Sb},\mathrm{Sn}\}$, written as:

$$\partial_t w_\alpha + c_\alpha \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} w_\alpha = 0\,, \qquad (17)$$

where the coupling is ensured by a boundary reflection operator $\mathscr{R}_{\mathbf{x}}$, $\mathbf{x} \in \partial\mathscr{D}$. The expression of its kernels $\rho_{\alpha\beta}(\mathbf{x},\hat{\mathbf{k}},\hat{\mathbf{p}})$, $\alpha,\beta \in E$, are again derived in [9], for either Dirichlet (clamped) or Neumman (free) boundary conditions for the underlying wave system. This study has shown that the shear mode T is uncoupled from the other modes on clamped or free boundaries, and the bending modes are uncoupled from the in-plane modes

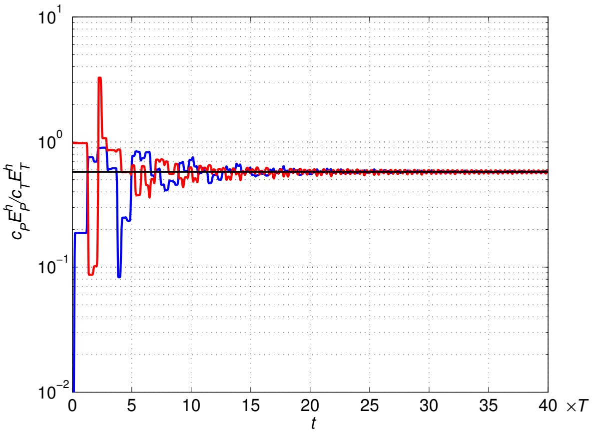

Figure 5. Evolution of the ratio $\mathscr{E}^h_{\mathrm{P}j}(t)/\mathscr{E}^h_{\mathrm{T}j}(t)$ in an assembly of two Timoshenko beams with $\phi = \frac{\pi}{4}$, $c_{\mathrm{T}2}/c_{\mathrm{T}1} = c_{\mathrm{P}2}/c_{\mathrm{P}1} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $T = L/c_{\mathrm{T}2}$, and a "hat" source with $\hat{k}_0 = +1$. 8-th order SSP Runge-Kutta DG method with 10-th order Legendre Gauss-Lobatto nodal expansion and $\mathscr{N} = 40$ elements, — beam #1 ($j = 1$), — beam #2 ($j = 2$), — $c_{\mathrm{T}j}/c_{\mathrm{P}j}$.

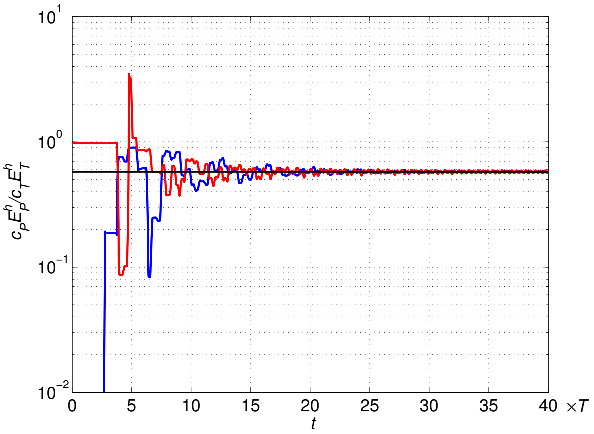

Figure 6. Evolution of the ratio $\mathscr{E}^h_{\mathrm{P}j}(t)/\mathscr{E}^h_{\mathrm{T}j}(t)$ in an assembly of two Timoshenko beams with $\phi = \frac{\pi}{4}$, $c_{\mathrm{T}2}/c_{\mathrm{T}1} = c_{\mathrm{P}2}/c_{\mathrm{P}1} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $T = L/c_{\mathrm{T}2}$, and a "square" source with $\hat{k}_0 = -1$. 8-th order SSP Runge-Kutta DG method with 10-th order Legendre Gauss-Lobatto nodal expansion and $\mathscr{N} = 40$ elements, — #1 ($j = 1$), — beam #2 ($j = 2$), — $c_{\mathrm{T}j}/c_{\mathrm{P}j}$.

as well. However mode conversions arise for the coupled bending modes Pb and Sb or in-plane modes Pn and Sn. The analysis underlines the existence of a critical incidence angle $\sin\theta_c = \sqrt{(1-\nu)/2}$ with respect to the boundary normal, corresponding to an incident transverse mode S reflected into a longitudinal mode P tangentially to the boundary. As mentioned above, this situation is presently ignored in our model.

The coupled transport equations (17) are solved numerically by the Runge-Kutta DG finite element method of section 3.2 partitioning the plate into 21×21 uniform quadrilateral elements, and using a third-order time integration scheme. A fifth-order Lagrange polynomial basis based on Gauss-Legendre interpolation points is considered for both dimensions in physical space, and a first order Fourier expansion is used for the discretization with respect to the phase variable $\theta \in [0, 2\pi[$ (such that $\hat{\mathbf{k}} = (\cos\theta, \sin\theta)^{\mathsf{T}}$, see section 3.2). Figure 7 displays our results for an isotropic Gaussian source:

$$w^0_{\mathrm{Pb}}(\mathbf{x}, \theta) = \exp\left(-\log 2 \frac{|\mathbf{x} - \mathbf{x}_0|^2}{R_0}\right)$$

of width $R_0 = 0.2 \times L$ and centered at $\mathbf{x}_0 = (+\frac{L}{2}, -\frac{L}{2})$, loading the bending mode Pb for example. Thus only the bending modes Pb and Sb propagate in this plate, however they get coupled once the energy rays have reached the boundaries. Poisson's ratio is $\nu = 0.3$, such that $\theta_c \simeq 36°$. The time scale is $T = L/c_{\mathrm{S}}$, the time needed by transverse rays to reach the boundary of the plate starting from its center. The Courant number for the simulations is fixed at $\mathrm{CFL} \equiv c_{\mathrm{P}} \frac{\Delta t}{h} = 10^{-1}$, $h = \min(\Delta x, \Delta y)$ the minimum side length of the quadrilateral elements. The overall mechanical energy computed from the numerical solutions $w^h_\alpha(\mathbf{x}, \theta, t)$ of (17) within the plate by:

$$\mathscr{E}^h(t) = \sum_{\alpha \in E} \int_{\mathscr{D}} \int_0^{2\pi} w^h_\alpha(\mathbf{x}, \theta, t) \mathrm{d}\theta \mathrm{d}\mathbf{x}$$

is equal to the total energy introduced into the system $\mathscr{E}(0) = 2\pi \int_{\mathscr{D}} w^0_{\mathrm{Pb}}(\mathbf{x}, \theta) \mathrm{d}\mathbf{x}$: thus it is verified that the proposed scheme in dimension two is numerically conservative. However it is rather dispersive due to its low resolution with respect to the phase variable θ. Increasing its Fourier order reduces significantly the dispersion error, but the positivity of the numerical solutions is not guaranteed. This condition gets satisfied using Lagrange interpolation [31], at the expense of a loss of accuracy by the ray effect. Yet these results illustrate the interest of imposing boundary conditions through the numerical flux in the DG method [12]. As to conclude on this example, it should be noted that the direct numerical simulations performed elsewhere [25, 26] using a Monte-Carlo method underlines the emergence of a diffusive regime at late times for plate or shell assemblies, as for the case of beam assemblies. This phenomenon should be also revealed by the DG scheme in (physical) dimension two.

5 CONCLUSIONS

A transport model for the vibrational energy evolution in heterogeneous slender structures has been proposed. The transport equations have been solved numerically by a Runge-Kutta discontinuous finite element method, among other possible schemes. This method has been implemented and tested on examples of thick beams or plates. Numerical simulations illustrate the transport regime for these structures, and the onset of a diffusive regime at late times. The latter is characterized by energy equilibration rules very much comparable to the assumption of modal equipartition invoked in the Statistical energy analysis of structural-acoustics systems [10]. However a rigorous mathematical model of the diffusive regime for bounded elastic media lacks at present. It should be able to predict the equilibration ratios as function of the sub-systems mechanical parameters. Another open issue is the consideration of the glancing set of energy rays–the diffractive and gliding ones–for the part of the power flows possibly trapped by the boundaries. Regarding the

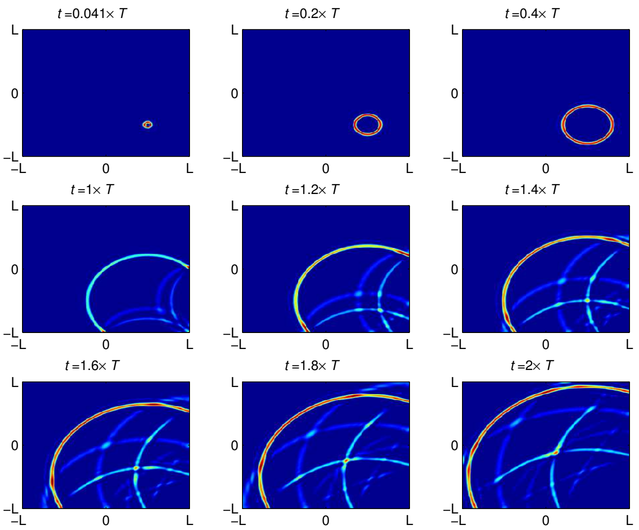

Figure 7. Evolution of the energy density in a flat plate for $\nu = 0.3$, $T = L/c_S$, and an isotropic Gaussian source. 3-rd order SSP Runge-Kutta DG method with 5-th order Gauss-Legendre nodal expansion in physical space and $\mathscr{N} = 21 \times 21$ elements, and 1-st order Fourier expansion in the phase variable.

DG method, further investigations are needed to improve the discretization schemes in order to enforce some positivity-preserving condition. These issues are the subject of ongoing research.

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